

# PENGYU WANG

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## EDUCATION

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| <b>University of Washington, Michael G. Foster School of Business</b><br><i>Ph.D. in Operations Management</i><br>Advisors: Prof. Shi Chen, Prof. Yong-Pin Zhou, Prof. Masha Shunko, Prof. Mingwen Yang | Seattle, WA, USA<br>Jan. 2023 – expected 2027     |
| <b>University of Washington, Department of Economics</b><br><i>Ph.D. in Economics</i> (program; transitioned to OM)                                                                                     | Seattle, WA, USA<br>Sept. 2020 – Dec. 2022        |
| <b>University of Wisconsin–Madison, Department of Economics</b><br><i>Master of Science in Economics</i>                                                                                                | Madison, WI, USA<br>Aug. 2018 – June 2020         |
| <b>Western University</b><br><i>Bachelor of Arts, Honors Specialization in Economics</i>                                                                                                                | London, Ontario, Canada<br>Sept. 2014 – June 2018 |

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## RESEARCH INTERESTS

**Topics:** Sustainable operations, Innovative value chain, Platform operations, AI infrastructure operations, Global supply chain challenges, Semi-conductor industry and supply chain.

**Methods:** Stochastic optimization, Game theory, Data analysis & Causal inferences, Mean-field games and control, Reinforcement learning.

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## RESEARCH

(†: Co-first authors).

### Working Papers

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1. **Wang, P.**, Chen, S., Hou, X. (2026). *Collective ESG Investment: Mitigation, Greenwashing, and Dynamic Strategic Interactions*. Reject & Resubmit at *Manufacturing & Service Operations Management* (Apr, 2026), Resubmitted (Aug, 2026).
  - Presented at INFORMS 2025.
  - \*Scheduled presentation at INFORMS 2026. **Date:** Tuesday, November 3, 4:15 PM-5:30 PM. **Room:** Moscone South-70 (Lower Mezz).
  - We build a dynamic stochastic framework to study how heterogeneous firms make collective ESG investment decisions, including greenwashing and substantive mitigation, when their actions are strategically interdependent and interact with an ESG-aware financial market.
2. Li, Y.<sup>†</sup>, **Wang, P.**<sup>†</sup>, Zhou, Y., Shunko, M. *Can Less Be More? A Hybrid Solution to Print-on-Demand Platforms*. Reject & Resubmit at *Manufacturing & Service Operations Management* (Nov, 2025)

- Accepted by and presented at MSOM conference 2024.
  - Studies hybrid make-to-stock / make-to-order (MTS/MTO) production and assortment regulation on print-on-demand platforms. Develops analytical models and contracts to align incentives between platforms and content creators.
3. **Wang, P.<sup>†</sup>**, Zhang, J.<sup>†</sup>, Chen, S., Yang, M. (2026). *Platform-Empowered Digital Finance and Traditional Investing: Spillover Effects through Financial Literacy and Risk Tolerance*. Under Review at *Manufacturing & Service Operations Management* (Aug, 2026).
    - \*Accepted by and scheduled talk at CIST conference 2026, and DSI conference 2026.
    - This paper studies whether digital finance substitutes for traditional channels or broadens market participation. Using individual-level investment data, we conceptualize digital finance as a gateway that reduces participation frictions, facilitates experiential learning, and promotes migration into adjacent financial markets.
  4. **Wang, P.**, Zhou, Y., Shunko, M. *Marketplace Governance and Production Strategy: Decentralized Assortment and Hybrid Fulfillment in Creator Economy*. In preparation for submission.
    - \* Scheduled talk at DSI 2026 conferece.
    - We study how creator portfolios are formed in digital marketplaces and the resulting production challenges for platforms. The separation between assortment creation and production decisions creates a need for marketplace governance, which platforms can address through market instruments that reshape creator portfolios and improve performance.

## Work in Progress

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5. **Wang, P.**, Chen, S. (2026) *A Two-Layer Hybrid Graph Mean-Field Game for Distributed LLM Inference Demand*.
  - Studies the offloading policy and pricing scheme in a data center network that provides AI-computing power.
6. **Wang, P.**, Chen, S. (2026) *Global Supply Chain Networks, Dynamics, and Disruption*.
  - Build a theoretical framework to uncover how global supply chain shocks penetrates through the supply network and how firms' endogenous R&D investments accelerate or contain such effects.
7. **Wang, P.**, Zhang, J., Chen, S. (2026). *Who Are the Winners? Business Opportunity Management in High-Tech Industry*.
  - Leverage a unique semi-conductor industry leading company's data to explore how such high-tech company explores business opportunity and foster future business partnership network.

## Proceedings

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8. **Wang, P.** (2017) *The Causal Relationship between Socioeconomic Status and Smoking Behavior: Evidence from Canada*. Proceedings in Advances in Economics, Business & Management.
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## CONFERENCES & INVITED TALKS

*A Two-Layer Hybrid Graph Mean-Field Game for Distributed LLM Inference Demand:*  
Decision Science Institute 2026, San Francisco, CA (scheduled)

*Marketplace Governance and Production Strategy: Decentralized Assortment and Hybrid Fulfillment in Creator Economy:*  
Decision Science Institute 2026, San Francisco, CA (scheduled)

*Collective ESG Investment: Mitigation, Greenwashing, and Dynamic Strategic Interactions:*  
INFORMS Annual Meeting 2026, San Francisco, CA (scheduled)  
POMS Annual Meeting 2026, Reno, NV  
POMS-HK 2026, Shenzhen, China  
INFORMS Annual Meeting 2025, Atlanta, GA

*Can Less Be More? A Hybrid Solution to Print-on-Demand Platforms:*  
INFORMS Annual Meeting 2024, Seattle, WA  
MOSM 2024 Annual Meeting 2024, Twin-City, MN

Others talks and experiences:  
MSOM 2026 Annual Meeting, Boston, MA  
SICS Annual Meeting 2025, Berkeley, CA  
INFORMS Annual Meeting 2023, Phoenix, AZ

## Society Services

Session chairs: INFORMS 2024, MSOM 2024, DSI 2026, INFORMS 2026

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## TEACHING EXPERIENCE

**Invited Instructor:** (Evaluations out of 5.0).

IGB 250 Winter 2026, Global Supply Chain: 4.5

IGB 250 Summer 2026, Global Supply Chain: 4.6

**Teaching Assistance:**

SCM 530A Autumn 2023, Operations Management And Process Analysis: 4.6

SCM 513 Autumn 2023, Managing And Mining Big Data: 4.5.

BUS AN 516 A/B Winter 2024, Operations Research Data Analytics: 4.2.

OPMGT 520A Spring 2024, Operations Management: 4.3

EMBA 533A Winter 2025, Operations Management And Systems Analysis: 4.5

QMETH 505A Spring 2025, Decision Modeling: 4.1;

SCM 520A Spring 2025, Managing Supply Chain Projects: 4.2.

## REFERENCES

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